# Xinran ZHAO

Phone: +852 66706439 | Email: ee.xrzhao@my.cityu.edu.hk | Personal Website: joyxr.github.io

## EDUCATION

- **City University of Hong Kong** *Aug. 2020 - Present*
  *Ph.D. in Electrical Engineering* Hong Kong SAR
  - Supervisor: Prof. Lin DAI
  - GPA: 4.15/4.00
- **Huazhong University of Science and Technology** *Sep. 2016 - Jun. 2020*
  *Bachelor in Electronics and Information Engineering* Wuhan, China
  - GPA: 3.86/4.0
  - Thesis: "Random access toward massive machine-to-machine communications: Performance analysis and protocol design", supervised by Dr. Yayu GAO.

## RESEARCH INTERESTS

- **Decentralized multiple access**: Modeling, analysis and optimal design.
- **Next-generation wireless communication networks**: Massive Internet of Things, low-latency communications, and distributed learning-based access design.

## PUBLICATIONS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION

**[S1]** Xinran Zhao and Lin Dai, "Throughput-optimal random access: A queueing-theoretical analysis for learning-based access design," submitted for publication in *xxx*. [Preprint]

**[J2]** Xinran Zhao and Lin Dai, "To sense or not to sense: A delay perspective," to appear in *IEEE Transactions on Communications*.

**[J1]** Xinran Zhao and Lin Dai, "Connection-based Aloha: Modeling, optimization, and effects of connection establishment," *IEEE Transactions on Wireless Communications*, vol. 23, no. 2, pp. 1008–1023, Feb. 2024.

**[C1]** Xinran Zhao and Lin Dai, "To sense or not to sense: A delay perspective," IEEE International Conference on Communications, Denver, CO, USA, 2024, pp. 2318-2323.

## TEACHING EXPERIENCE

- **Tutor for Final Year Projects**
  - Optimal Backoff Tuning for Massive Access of Machine-type Devices in 5G Networks *Fall 2024*
  - Massive Access for Beyond-5G Communications Networks: Connection-based or Connection-free? *Fall 2021, Fall 2022, Spring 2024*
  - Energy-Efficient Massive Access for Machine-to-Machine Communications *Fall 2020*
- **Teaching Assistant**
  - Data Engineering and Learning Systems *Spring 2024*
  - Mobile Communication and Networks *Spring 2022, Fall 2022, Fall 2023*
  - Principles of Communications *Fall 2021, Spring 2023*

## AWARDS

- Research Tuition Scholarship *City University of Hong Kong, Sep. 2023*
- EE Graduate Research Seminar Award - 3nd Prize *Department of Electrical Engineering, City University of Hong Kong, May 2022*
- Outstanding Undergraduate Thesis Award *Huazhong University of Science and Technology, Jun. 2020*

## PROJECTS

- **Intelligent Control System of Tobacco Baking Room** *Mar. 2019 - Jan. 2020*
  *Team leader of a collaboration project between Dian Group (student tech team in HUST) and China Tobacco* Wuhan, China
  - Developed a **back-end** system based on Python Django, MySQL and Redis. The implemented functions included receiving and storing the data reported by IoT terminals in baking rooms, monitoring the status of baking rooms, historical data statistics, remote control, etc.
  - Conducted the joint debugging between the server and IoT embedded terminals, the communications between which were implemented based on the **LoRaWAN** architecture.
  - Deployed the system in practical tobacco-baking scenarios, which enables automatic baking and monitoring.
- **Online Exam System** *Sep. 2018 - Jan. 2020*
  *Member of a collaboration project between Dian Group* Wuhan, China
  - Developed a **back-end** system based on Python Flask and MySQL. The implemented functions included the management of question bank and examinee information, online examination, grading, etc.
  - Applied the system for Interior Design Qualification Exams.

## SKILLS

- **Programming Languages:** Python, MATLAB, C.
- **Mathematical Tools:** Probability theory, queueing theory, stochastic process, statistical inference.
- **Web Technologies:** Python Flask, Python Django, MySQL, Redis.

## ADDITIONAL INFORMATION

- **Languages:** Mandarin (native) and English (IELTS 7.0)